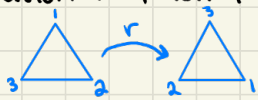
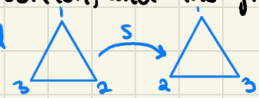
Jan. 29

- Continued worksheet from previous class, made changes to front side
- Notice notation, ex. $r \circ s \circ r \circ r = rsrr$ can get cumbersome, so we can use exponents, i.e $= rsr^2$

when we write groups multiplicatively, exponential notation is nice

- The symmetries of a triangle form a group
  - The group operation is function composition, and the group has six elements

if we label 

and 

the elements are

$\{e = id, r, r \circ r = r^2, s, rs, r^2 s\}$

- This group is called the Dihedral Group on 3 vertices.
  - We denote it by D_3 ★Note, some authors write D_6 for # of elements

    # of vertices

Generators & Relations

$D_3 = \langle r, s : r^3 = s^2 = e, \text{ and } rs = sr^2 \rangle$

D_3 is generated by symbols r & s subject to the relations $r^3 = s^2 = e$ and $rs = sr^2$

- D_3 is not Abelian. In particular, $rs \neq sr$

  commutative

- Thm. 2.4 (The Socks-Shoes Property) Let a, b be elements of a group G. The inverse of ab is the inverse of b group operationed with the inverse of a. Written symbolically, let $a, b \in G$. Then $(ab)^{-1} = b^{-1}a^{-1}$.

  Let's use a superscript -1 to denote the inverse, so $(ab)^{-1}$ is the inverse of ab

  Proof We compute

  $b^{-1}a^{-1} * ab = b^{-1}(a^{-1}a)b = b^{-1}eb = b^{-1}b = e$. Thus, $b^{-1}a^{-1}$ functions as an inverse to ab. We have proved that inverses are unique, thus $b^{-1}a^{-1}$ is the inverse of ab. i.e $b^{-1}a^{-1} = (ab)^{-1}$. ★We should really prove it is a right inverse too, but it will turn out proving either one suffices. We'll prove this later, with some more work.

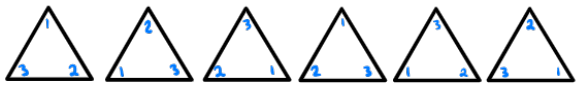
Dihedral Groups: The Example of D_3

Math 470: Abstract Algebra

Put the triangle like this △. For simplicity, make sure the vertices are labeled like the one on the board. Pick the triangle up, do some transformation, and put it back where it was. (Put it on the same "footprint" but with potentially different vertices in different places.)

1) How many different positions can the triangle end up in?

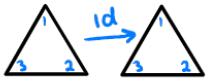
6 different positions



2) Shift from thinking about the positions to thinking about the transformations that get you from one position to another. We often call these transformations *symmetries*. We think of two transformations as being the same if they get the triangle from the same starting point to the same ending point. Come up with a scheme for labeling the transformations, and write all your transformations down. (Talk to the instructor before continuing on.)

We can:

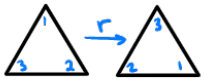
- do nothing (id)



- rotate again ($r_{240} = r \circ r$)

- rotate once (r)

120°



- flip (s)

3) The transformations are functions from the triangle to itself. We can define a binary operation on the transformations by composing them using function composition. Construct an operation table or Cayley table for the transformations. (Each row will be a labeled with a transformation element as will each column. The entry will be given by the composition of the row and column, with the row acting first.)

blank squares weren't done in class and I don't trust myself not to misinform the people ¯_(ツ)_/¯

$r^2 = r \circ r = R_{240}$

encouraged to try them yourself! •▽•

	e	r	r^2	s	$s' = sr$	$s'' = rs$
e	e	r	r^2	s	s'	rs
r	r	r^2	e	$s \circ r = s'$	$s' \circ r = s'' = rs$	$s'' \circ r = s$ $rs \circ r$
r^2	e	r	$s \circ r^2 = s'' = rs$	$s' \circ r^2 = sr\,r^2 = s$	$s'' \circ r^2 = rsr^2 = srr^2 = sr = s'$	
s	s	$r \circ s = s''$	$r^2 \circ s = sr = s'$			
s'	s'					
$rs = s''$	rs					

"row acting first" puts row operation on the right

column ∘ row = column(row)

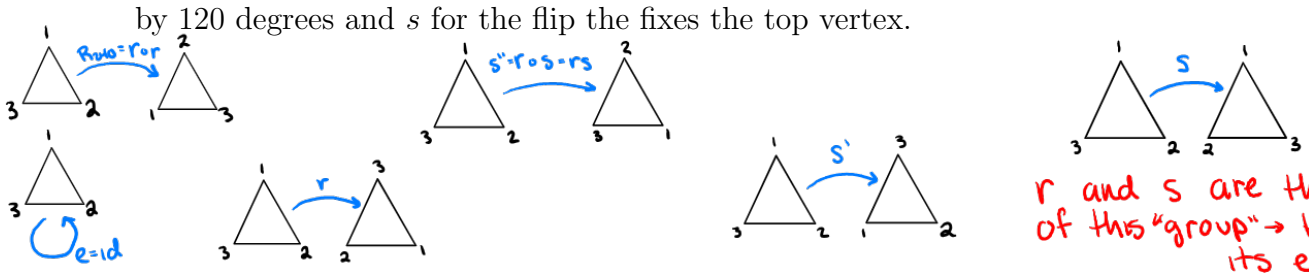
i.e $f \circ g = f(g)$

$f(x) \circ g(x) = f(g(x))$

note $sr \neq rs$, so this "group" is not Abelian

A group is Abelian if the table is symmetric across the diagonal

4) Describe all the transformations in terms of just two. I like to use r for rotation clockwise by 120 degrees and s for the flip the fixes the top vertex.



r and s are the generators of this "group" → they form all its elements

5) Prove the symmetries of a regular triangle form a group. (For this class, you always may assume that function composition is associative.) We call this group the dihedral group of order six and denote it D_3. (The 3 indicates the number of vertices of the polygon.) Is this group abelian?

<u>Verify the group conditions</u>

1) Associativity

Given in instructions, function composition is associative, i.e $(r \circ s) \circ r = r \circ (s \circ r)$, etc

2) Identity

The identity symmetry, i.e not moving the triangle, acts as the identity element.

3) Inverses

for s, it is shown that $s \circ s = id$, since flipping twice returns the triangle to the original position. So s^{-1} exists, namely s itself. for r, $r \circ r_{240} = r \circ r \circ r$ returns the triangle to the original position ($120° + 120° + 120° = 360°$ rotation) so r^{-1} exists, namely $r_{240} = r \circ r$. So, every element (symmetry) has an inverse.

Thus, the symmetries of a regular triangle form a group.

6) Repeat what you did for the triangle with the square □.

7) You can do this for any regular polygon! List some things you can already surmise about D_n.